

S7CAS-PLEIADES-CARTE-MERE

Pleiades

01

Revision 1.00

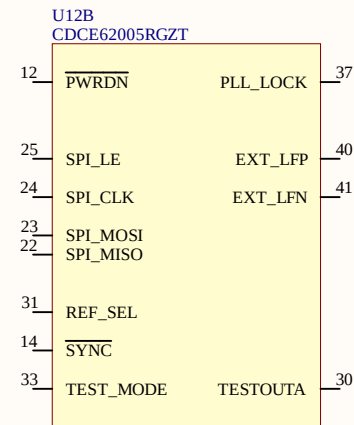
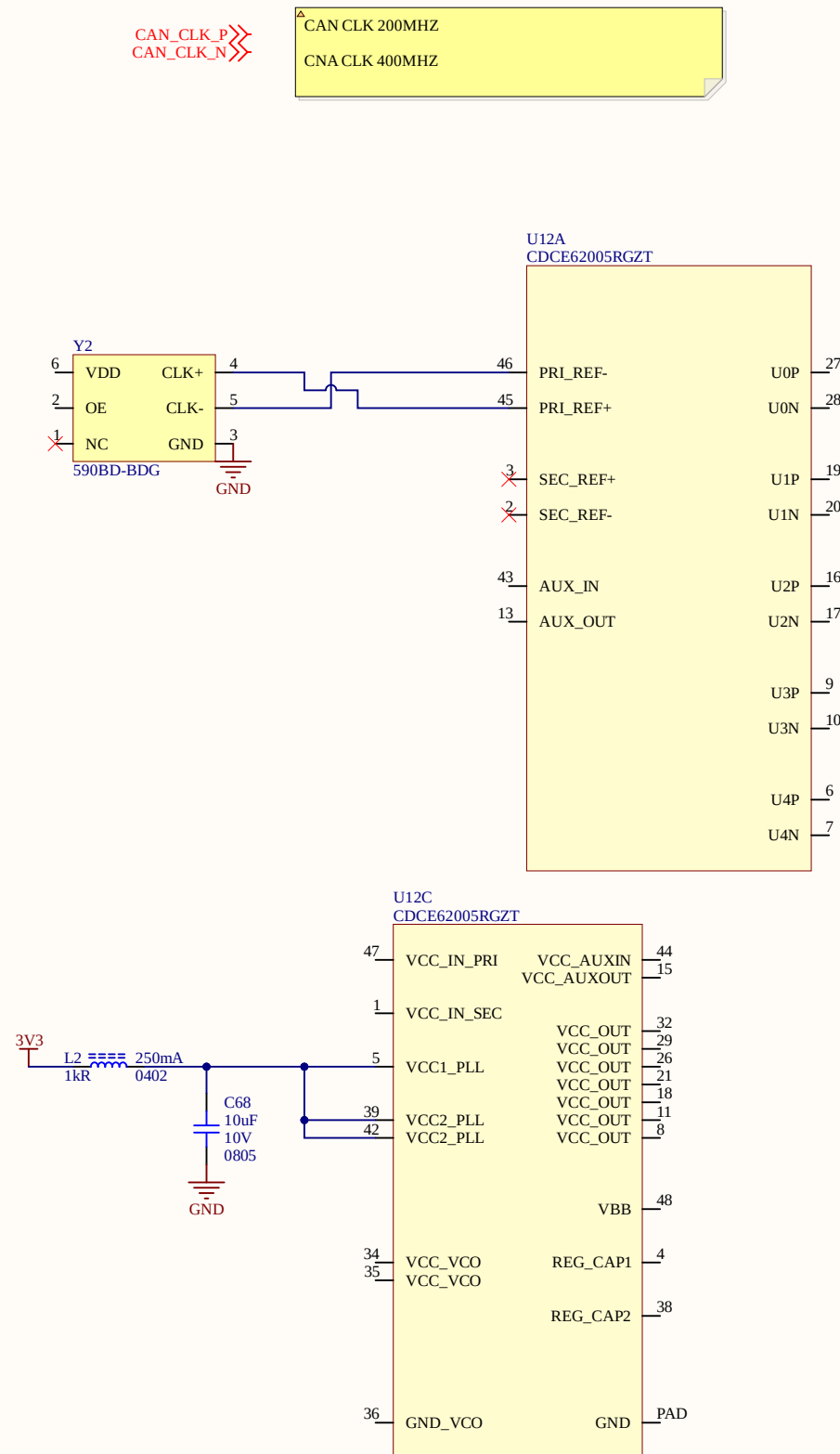
Date : 13/01/2026

Revision history	

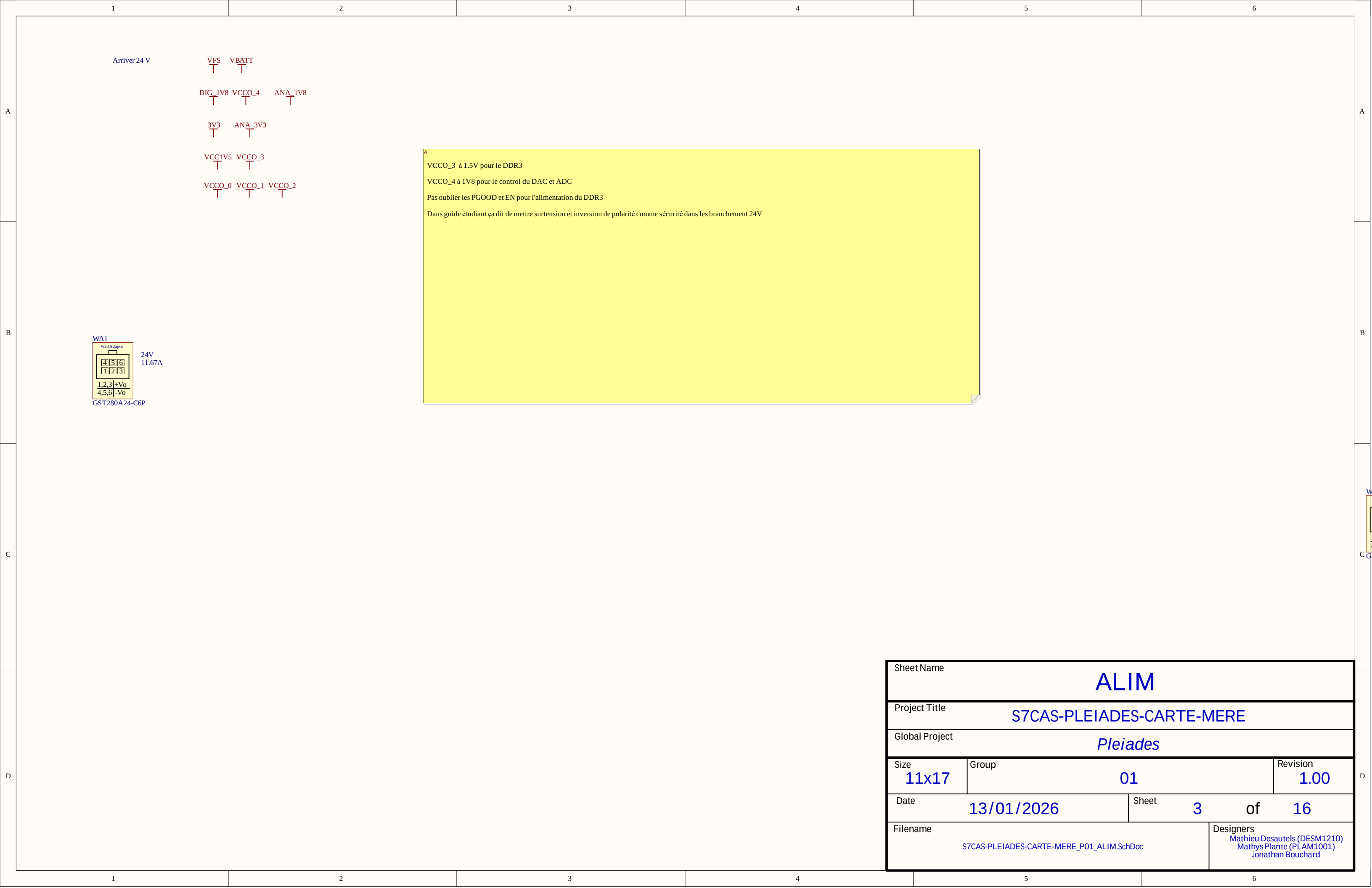
Package size conversion	
Metric	Imperial
1005	0402
1608	0603
2012	0805
3216	1206
3225	1210
6432	2512



Project Title			
S7CAS-PLEIADES-CARTE-MERE			
Global Project			
Pleiades			
Size	Group		Revision
11x17	01		1.00
Date	Sheet		
13/01/2026	1 of 16		
Filename		Designers	
S7CAS-PLEIADES-CARTE-MERE_P01_TITLE.SchDoc		Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard	

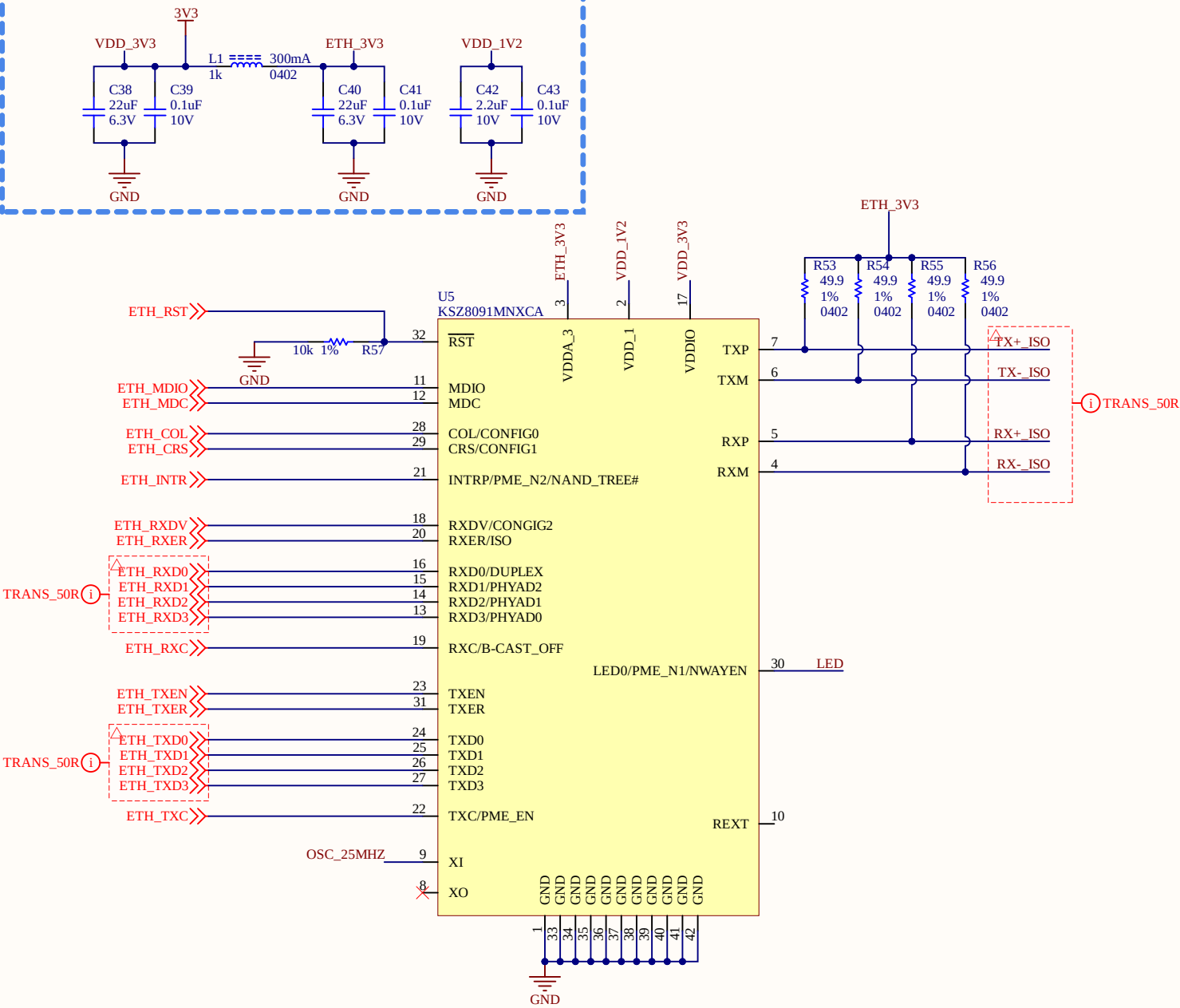


Sheet Name			CLOCK		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	Group		Revision		
11x17	01		1.00		
Date	13/01/2026		Sheet	2	of 16
Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_CLOCK.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		



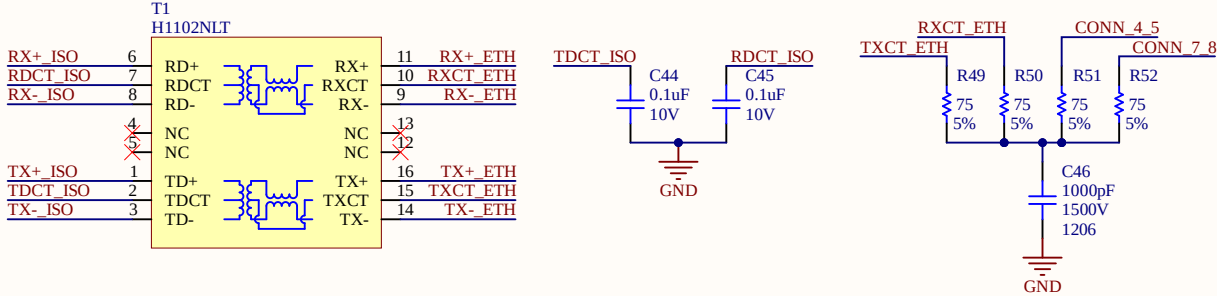
ETHERNET
ALIMENTATION

ETHERNET - PHY - KSZ8091MNXCA.pdf

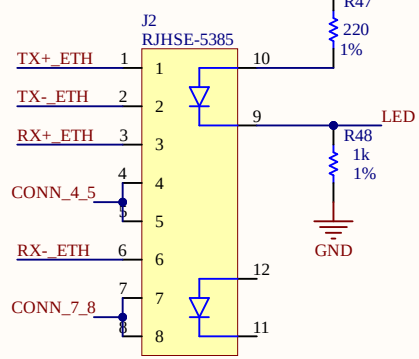


Ferite de 1k?
Valider les condensateur (Surtout pour ETH_3V3)
Faut tu mettre transmission entre Iso et Connecteur

ISO



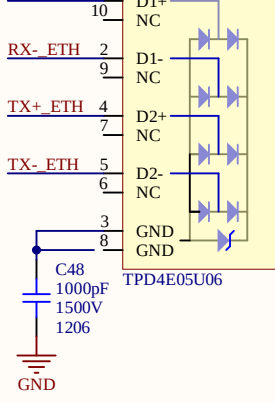
CONNECTEUR



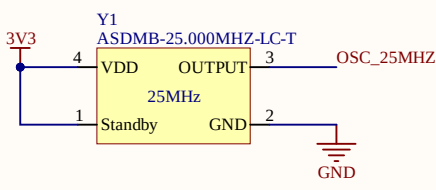
Pin #	Wire Color Legend	Signal
1	White/Green	TX+
2	Green	TX-
3	White/Orange	RX+
4	Blue	TRD2+
5	White/Blue	TRD2-
6	Orange	RX-
7	White/Brown	TRS3+
8	Brown	TRD3-

-Isolation du ETHERNET 1.5kV
-Ajout d'un condensateur de 1.5kV
-Ligne de transmission de 50R

ESD

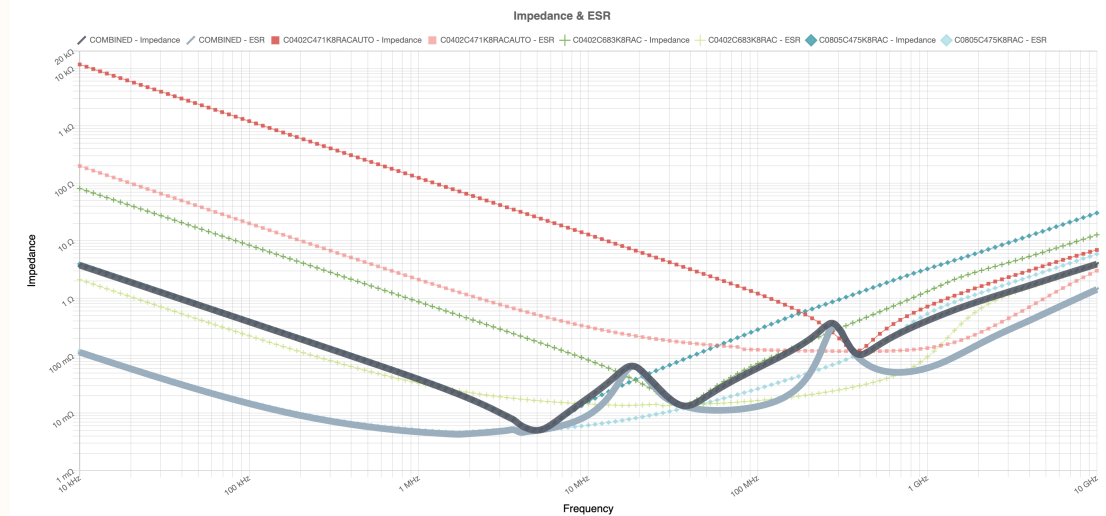
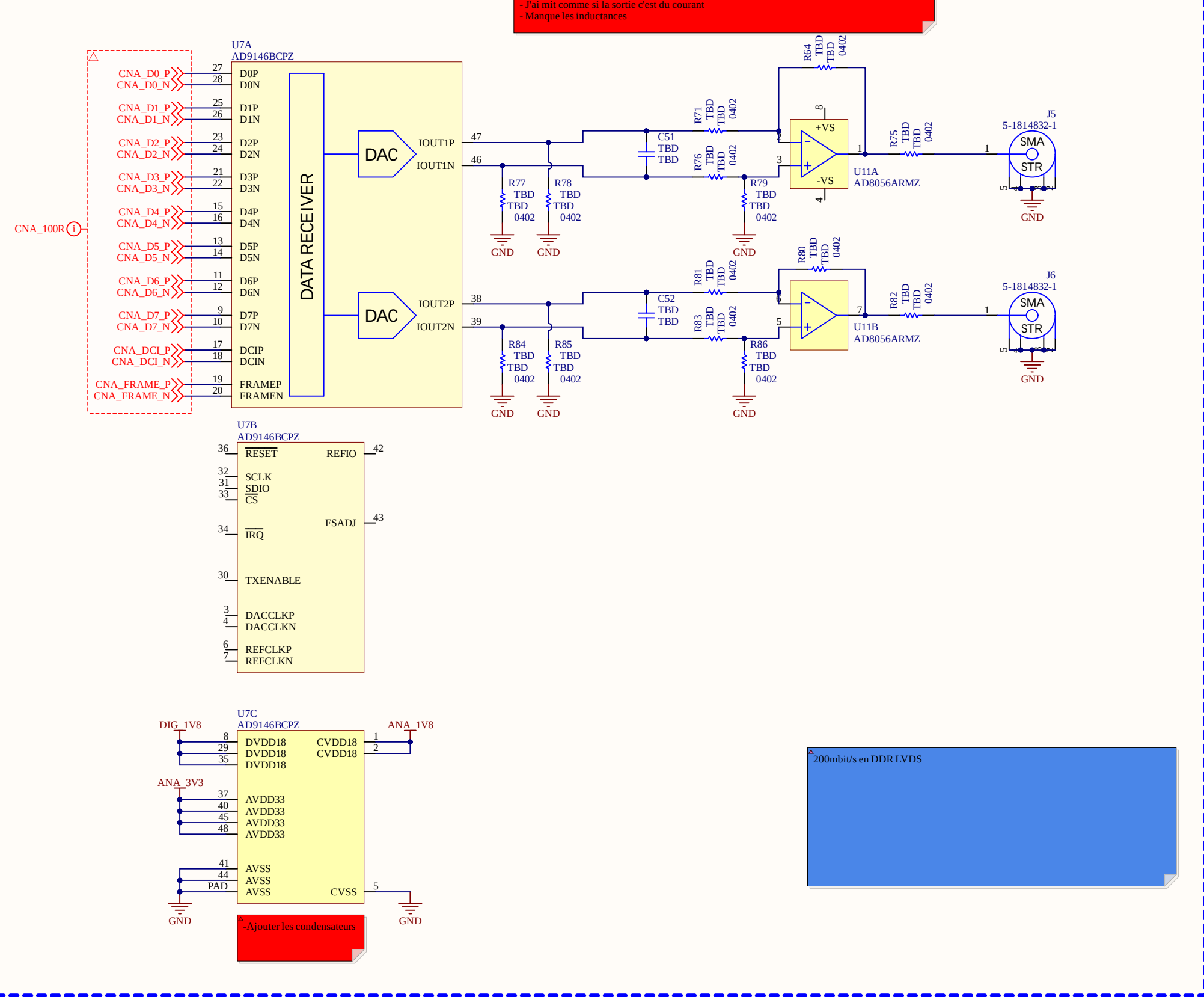


OSCILLI

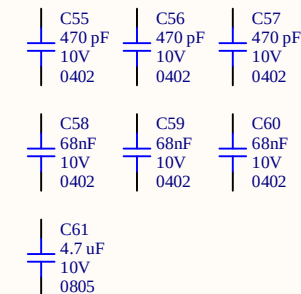


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Global Project			Pleiades		
Size	Group			Revision	
11x17	01			1.00	
Date	13/01/2026		Sheet	4 of 16	
Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_ETHERNET.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

CNA (DAC)



Legend	Tolerance	Part Number	Cap.	V _{DC} *	Dielectric	Qty.	Bias (V)	Amb. (°C)	Add	Remove	
-			Combined	4.3 uF	10	X7R	1	0	25		X
	10% ▾	C0402C471K8RACAUTO	470 pF	10	X7R	3	0	25	+	X	
+	10% ▾	C0402C683K8RAC	68 nF	10	X7R	3	0	25	+	X	
	10% ▾	C0805C475K8RAC	4.7 uF	10	X7R	1	0	25	+	X	



200mbit/s en DDR LVDS

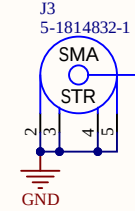
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Global Project			Pleiades		
Size	Group	Revision			
11x17	01	1.00			
Date	13/01/2026	Sheet	5	of	16
Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_CNA.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

CAN (ADC)
DECOUPLAGE

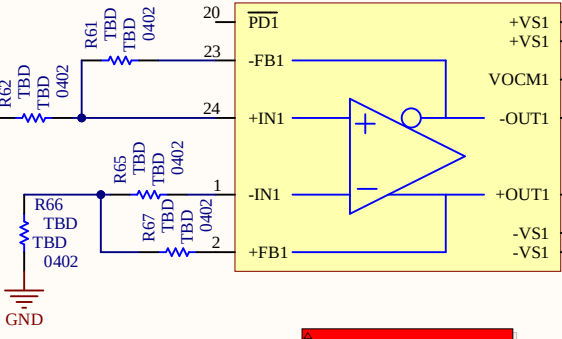
OPAMP - ADA4927-1_ADA4927-2.pdf
P.20

Ajuster les valeurs des résistances

Manque les alims et inductances

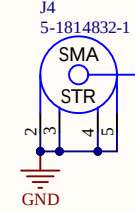


Valider si toute les autres pins Ground

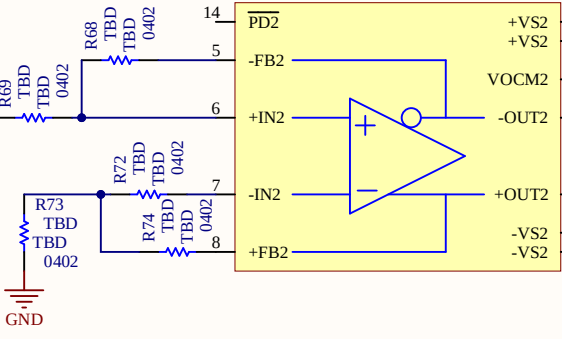


Ajuster les valeurs des résistances

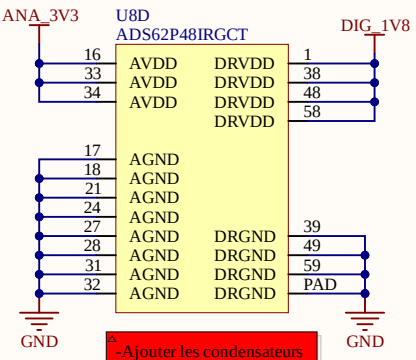
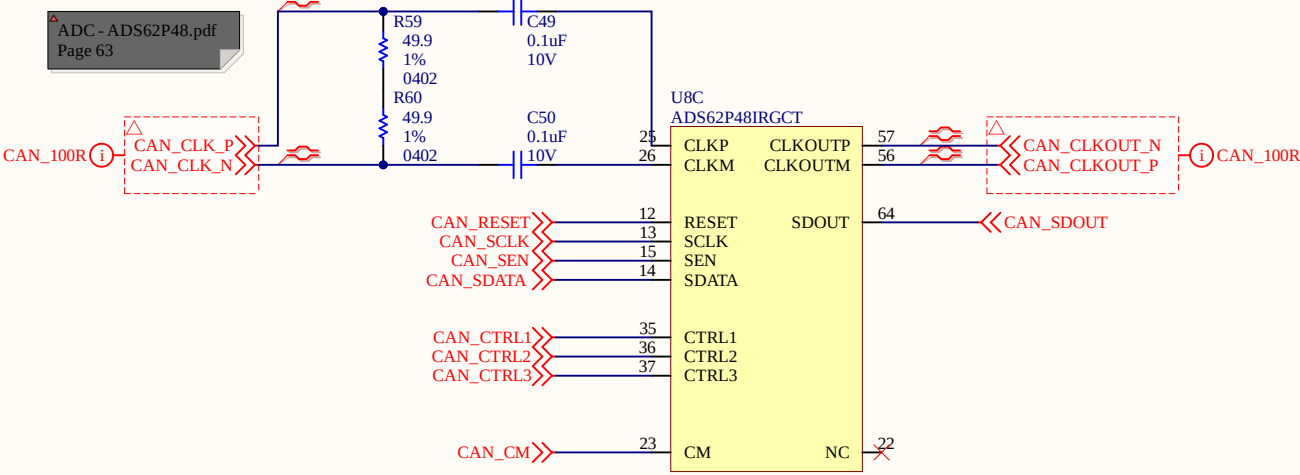
Valider si c'est 50ohms différentiel?



Valider si toute les autres pins Ground

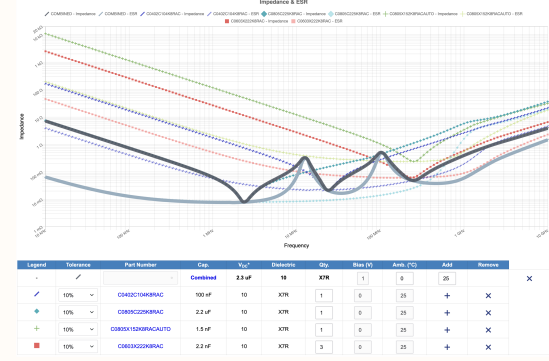


ADC - ADS62P48.pdf
Page 63



Ajouter les condensateurs

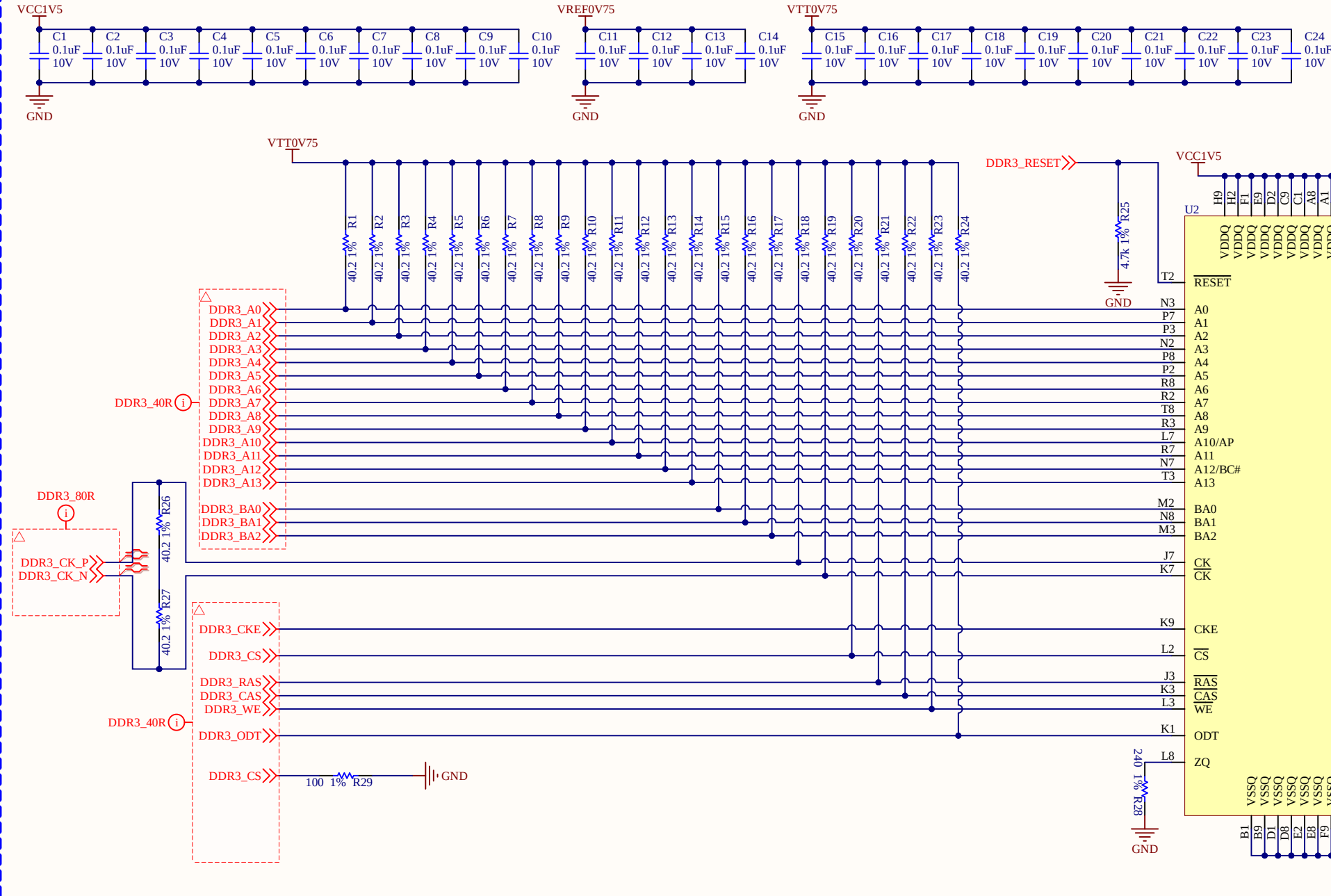
-Choix de OP-AMP pour isolé et pour passer d'analog a digital



C62	100 nF	C63	2.2 uF	C64	1.5 nF	C65	2.2 nF	C66	2.2 nF	C67	2.2 nF
10V	0402	10V	0805	10V	0805	10V	0603	10V	0603	10V	0603

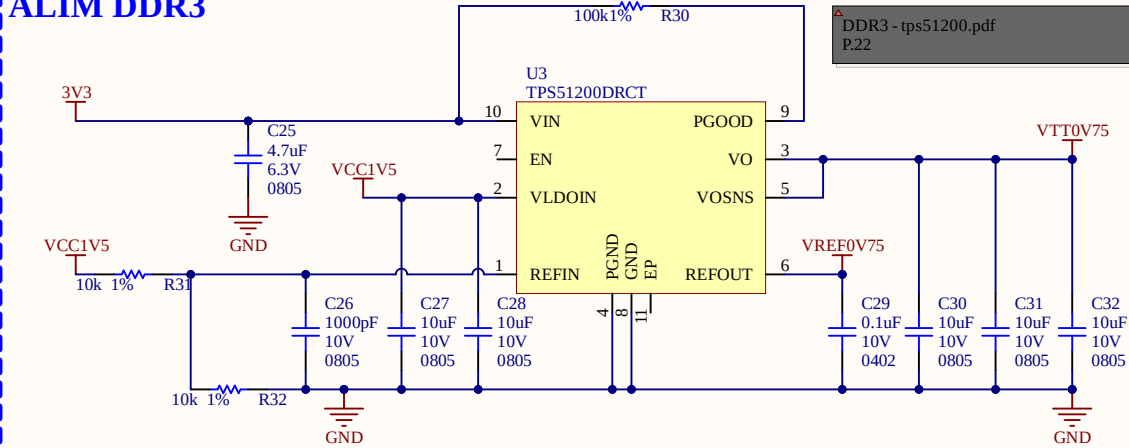
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Global Project			Pleiades		
Size	11x17	Group	01		Revision
Date			13/01/2026		1.00
Date			Sheet	6	of 16
Filename			S7CAS-PLEIADES-CARTE-MERE_P01_CAN.SchDoc		Designers
					Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard

FPGA



MANQUE ZIO?
ANALYSER LES COURANTS POUR LE VTT
AJUSTER LES CONDENSATEURS

ALIM DDR3

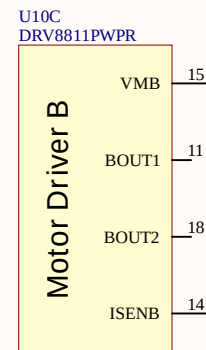
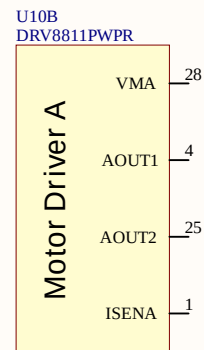
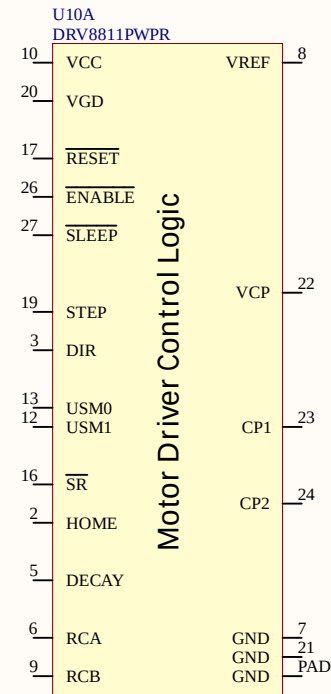


DDR3 - tps51200.pdf
P.22

-Ligne de transmission de 40R
-Resistance parallèle sur la ligne de transmission de 40.2Ohms
-Clock en pair différentiel 80R

Sortie PGOOD?
EN?

Sheet Name		DDR3	
Project Title		S7CAS-PLEIADES-CARTE-MERE	
Global Project		Pleiades	
Size	Group	Revision	
11x17	01	1.00	
Date	13/01/2026	Sheet	7 of 16
Filename		Designers	
S7CAS-PLEIADES-CARTE-MERE_P01_DDR3.SchDoc		Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard	

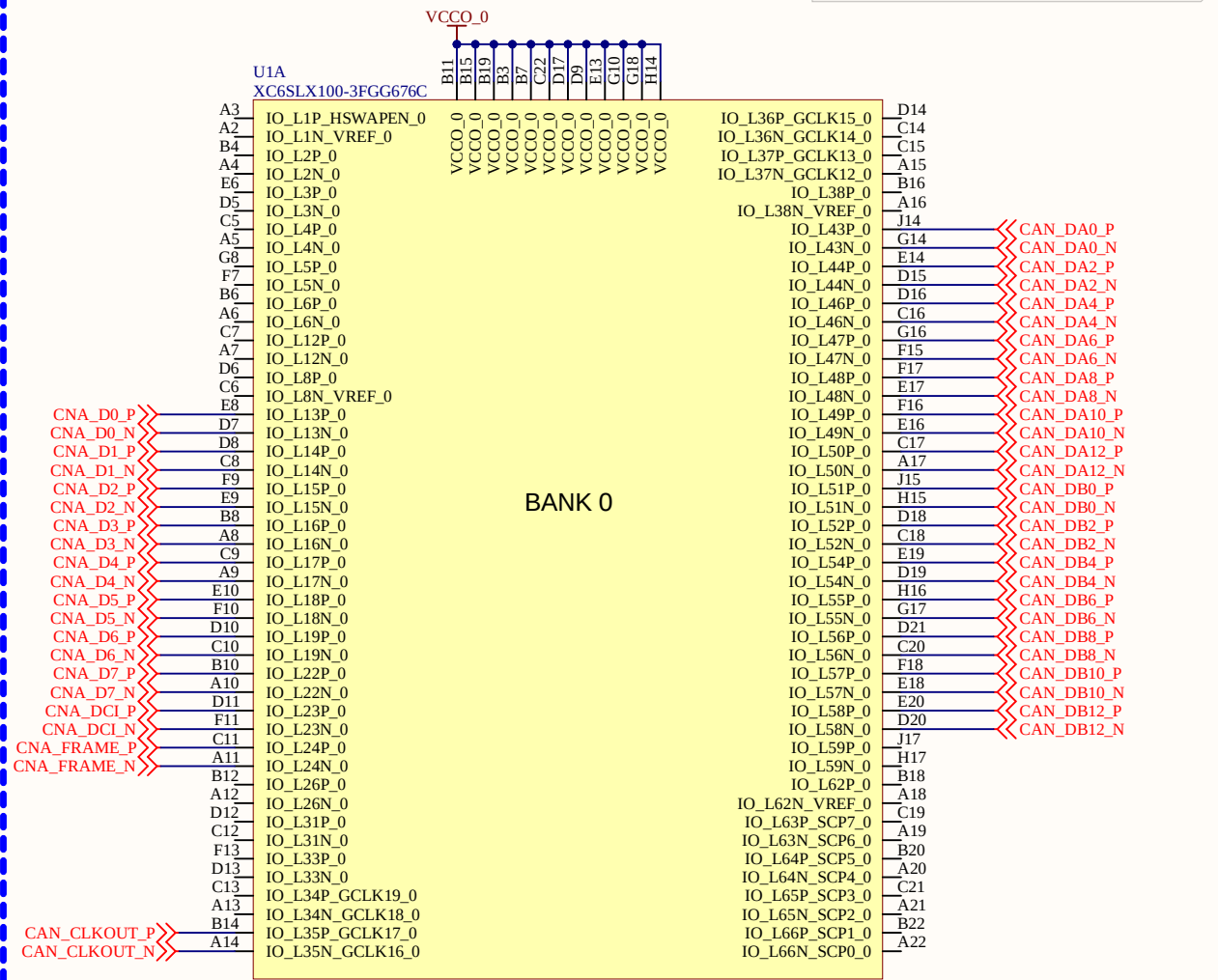


Sheet Name			MOTORS		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	Group		Revision		
11x17	01		1.00		
Date	13/01/2026		Sheet	8	of 16
Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_MOTORS.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

1	2	3	4	5	6
A					
B					
C					
D					

Sheet Name		ASIC	
Project Title		S7CAS-PLEIADES-CARTE-MERE	
Global Project		Pleiades	
Size	Group	Revision	
11x17	01	1.00	
Date	13/01/2026	Sheet	9 of 16
Filename		Designers	
S7CAS-PLEIADES-CARTE-MERE_P01_ASIC.SchDoc		Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard	

FPGA

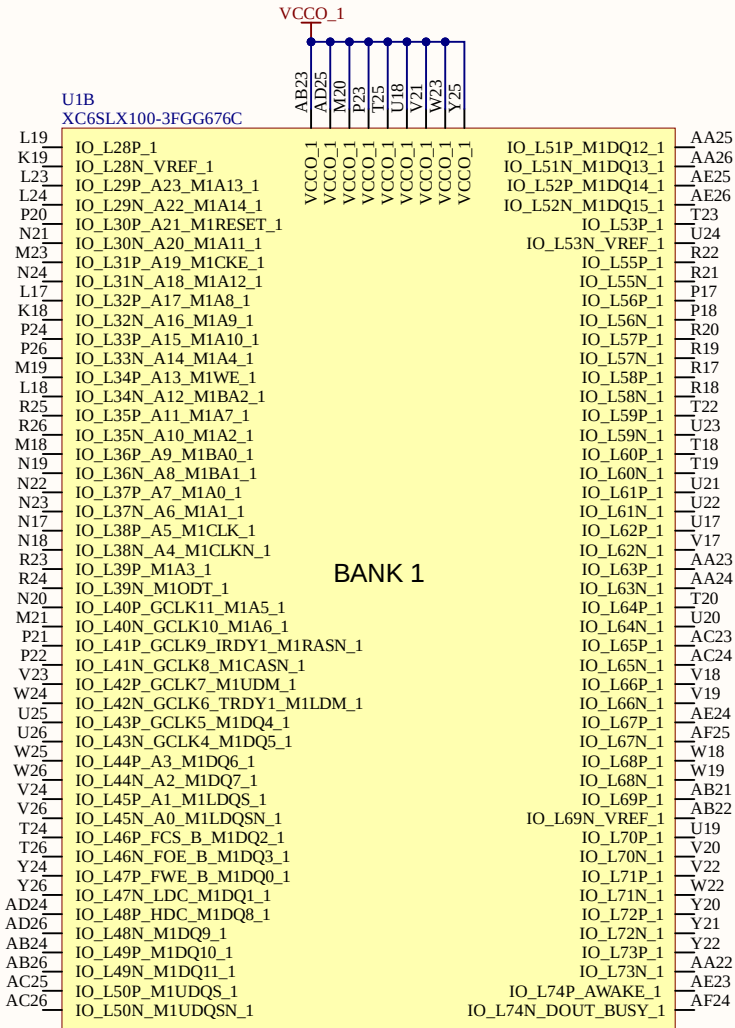


IO CAN est LVDS donc il faut DIFF_TERM = "TRUE"

- LVDS seulement sur BANK0 et BANK2

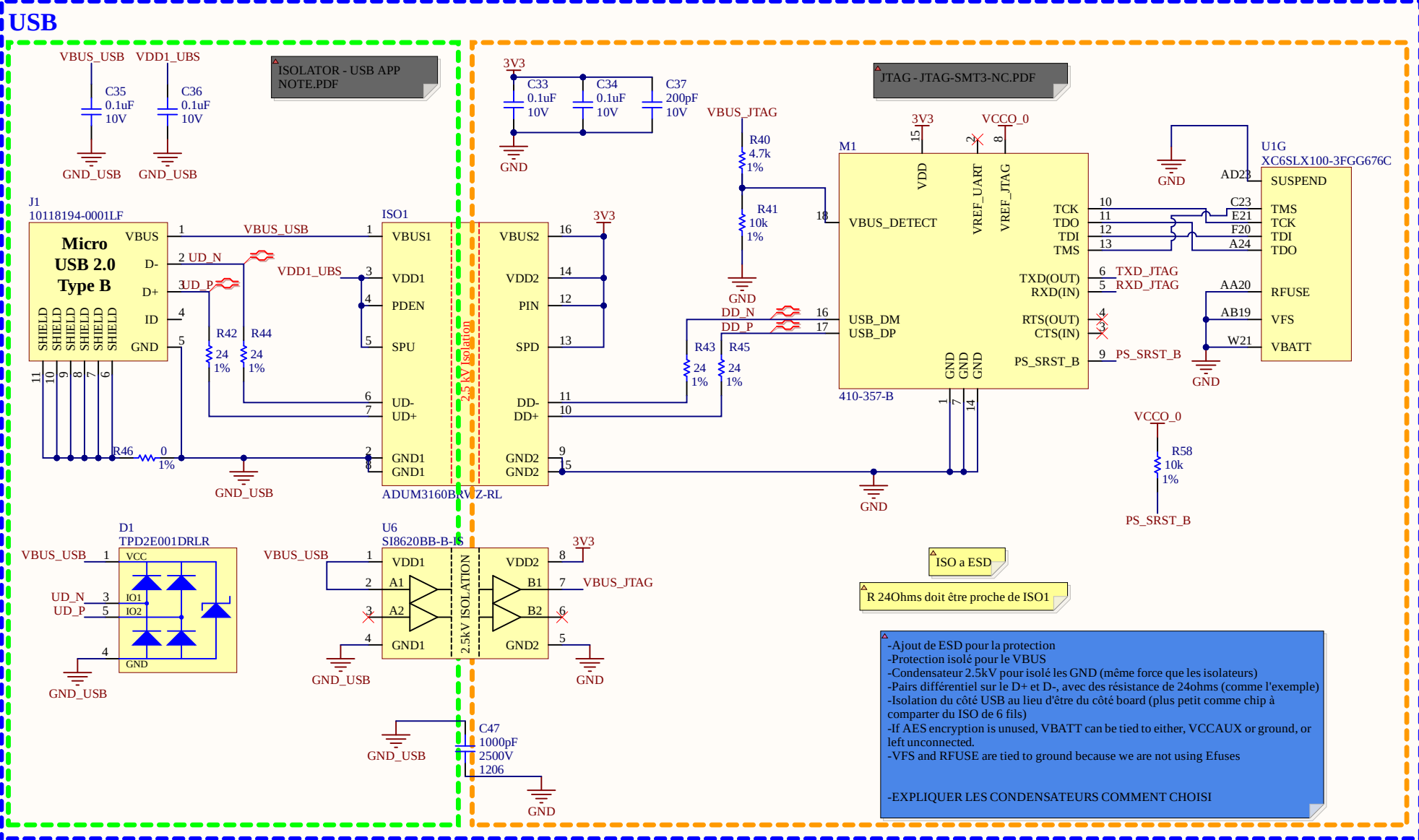
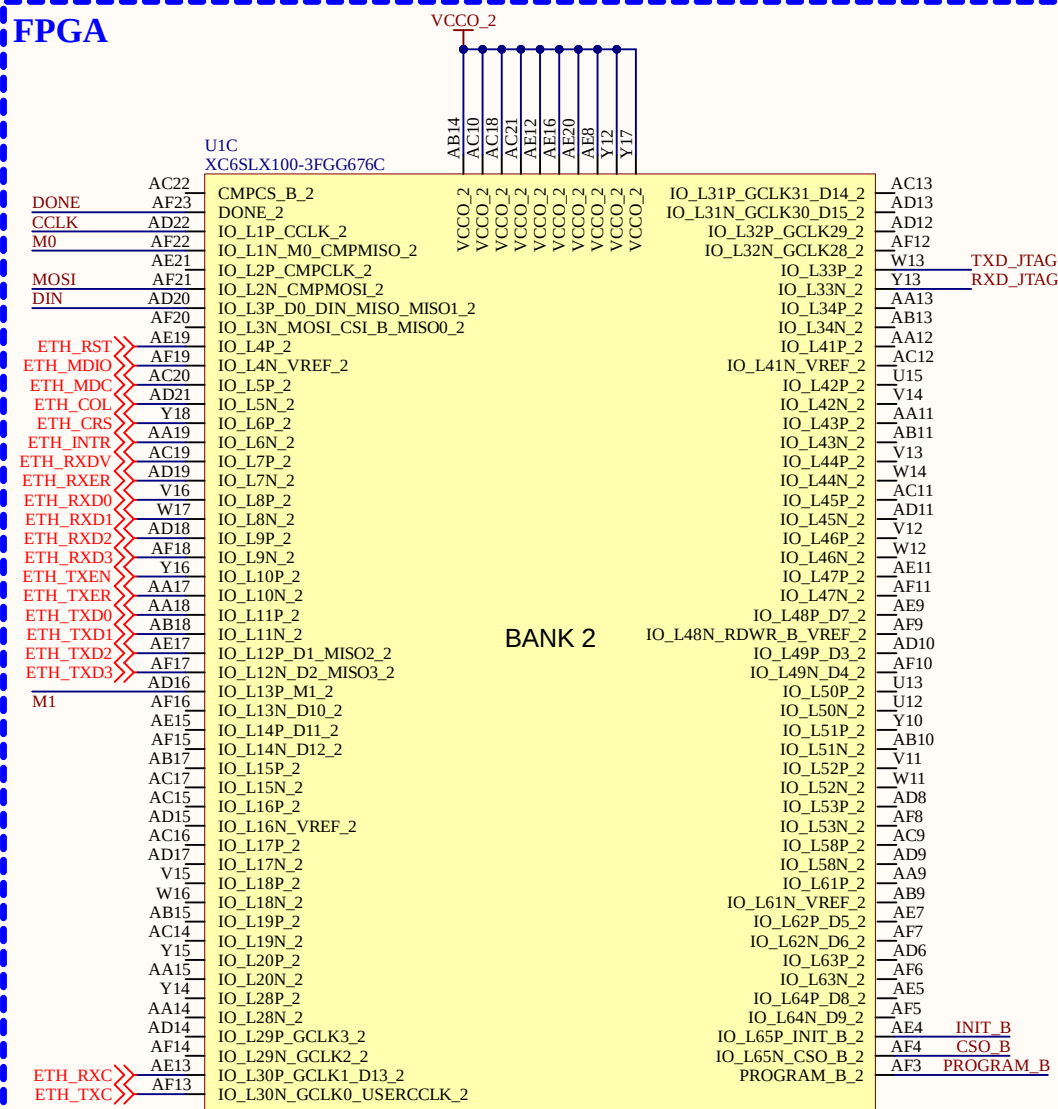
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Global Project			Pleiades		
Size	Group			Revision	
11x17	01			1.00	
Date			Sheet		
13/01/2026			10 of 16		
Filename				Designers	
S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK0.SchDoc				Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard	

FPGA

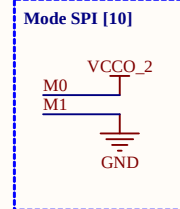
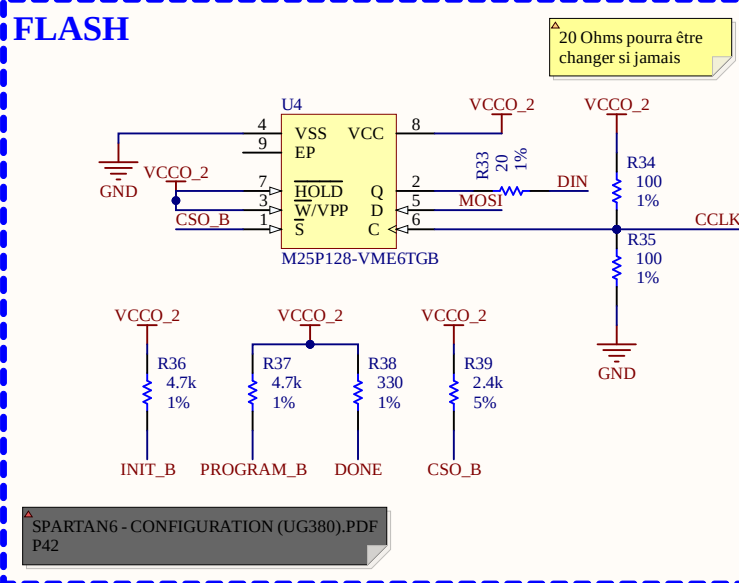


Sheet Name			FPGA - BANK 1		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	Group			Revision	
11x17	01			1.00	
Date	13/01/2026		Sheet	11	of 16
Filename				Designers	
S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK1.SchDoc				Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard	

FPGA



FLASH

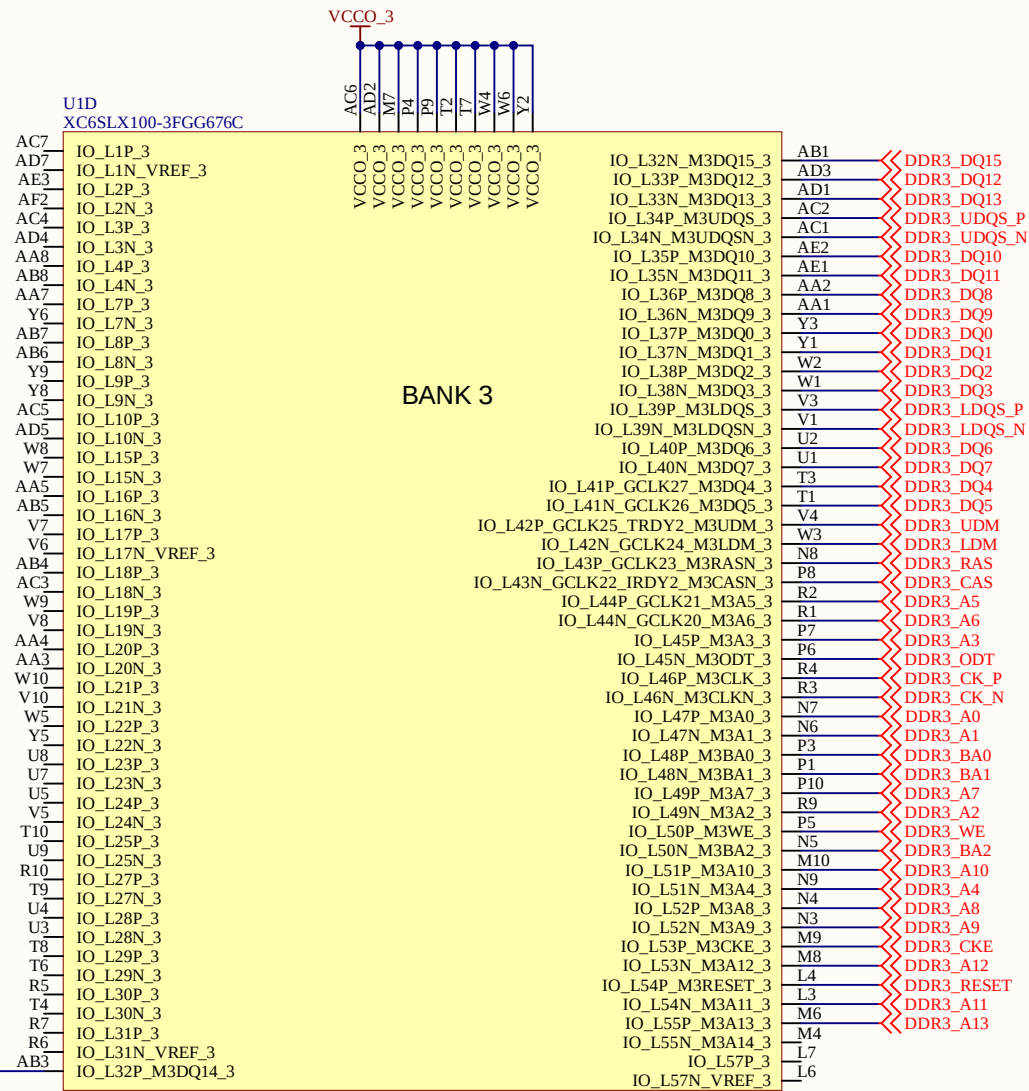


- M0 et M1 choisi pour avoir le mode SPI
- Ajout d'une résistance de 20 ohms sur Q

- Ajuster les condensateurs
- Pas sur si PS_SRST_B c'est obligatoire

Sheet Name			FPGA - BANK 2		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	Group		Revision		
11x17	01		1.00		
Date	13/01/2026		Sheet	12	of 16
Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK2.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

FPGA

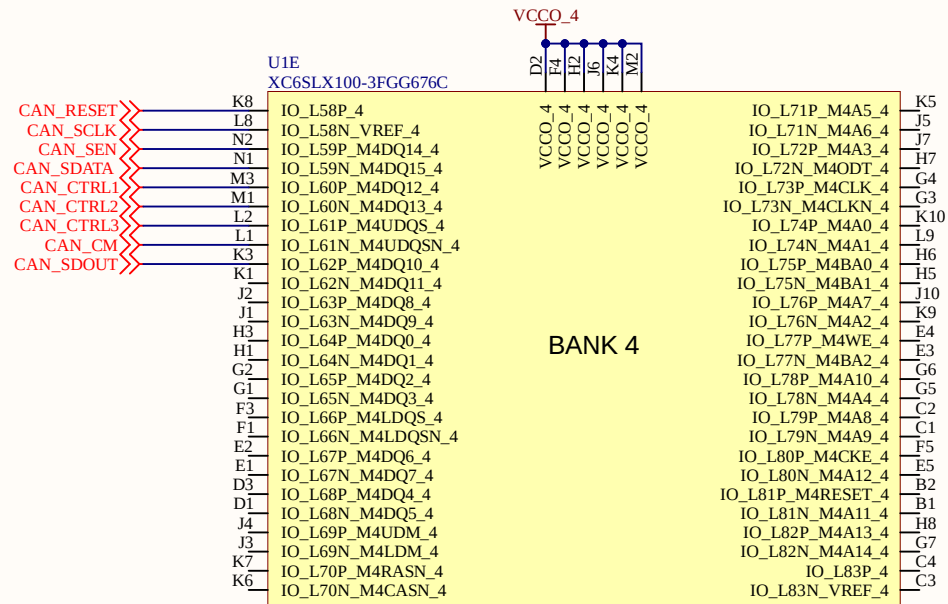


Rajouter les VREF avec des condensateurs pour le FPGA

VREF0V75

Sheet Name			FPGA - BANK 3		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	Group		Revision		
11x17	01		1.00		
Date	13/01/2026		Sheet	13	of 16
Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK3.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

FPGA



Sheet Name			FPGA - BANK 4		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	Group	Revision			
11x17	01	1.00			
Date	13/01/2026	Sheet	14	of	16
Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK4.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

